

Quantum-Inspired Hybrid Neural Networks for Neural Decoding: A Controlled Ablation Study of Learnable Quantum Sidecar Integration

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Abstract. We study parameterized quantum circuits (PQCs) integrated as residual sidecar modules within a ResNet-50 backbone for 31-class neural population decoding-imagined handwriting classification from multi-neuron spike rasters. Under strictly controlled conditions (fixed data splits, seeds, and optimizer), we compare four model variants: baseline, quantum sidecar with frozen input projection, quantum sidecar with backbone gradient-trained projection, and a measurement-guided variant that aligns angle encodings with circuit measurement outcomes. The backbone-gradient variant improves accuracy in 3/4 seeds (+0.19% mean, 95% CI [−1.10%, +1.48%]) and consistently reduces Linear CKA similarity to baseline features ($\Delta = -0.025$, 4/4 seeds), indicating genuine structural reorganization of representations. A nine-variant ablation identifies simple shallow architectures as the most effective and reproducible configuration. Measurement-guided training consistently improves representation geometry without reducing accuracy. All results use noiseless statevector simulation on 4 qubits, a regime chosen to reflect the practical constraints of current near-term superconducting hardware; no quantum computational advantage over classical methods is claimed.

Keywords: Neural decoding · Brain-machine interfaces · Hybrid neural networks · Representation geometry · Quantum machine learning

1 Introduction

Neural population coding research has established that brain-machine interface (BMI) systems benefit not only from decoding accuracy but from the geometric

quality of learned representations [9]. Deep networks trained on neural recordings may achieve high accuracy while learning geometrically suboptimal feature spaces for cross-session robustness.

Parameterized quantum circuits (PQCs) implement topology-constrained global feature mixing through quantum entanglement—an architectural prior structurally distinct from learned correlation in MLPs [8,1]. We ask whether a small, shallow PQC sidecar produces measurable, reproducible effects on representation geometry when integrated into a strong classical backbone under controlled conditions.

We evaluate a 4-qubit residual sidecar on 31-class neural decoding of imagined handwriting characters [2], using fixed data splits, initialization, and optimizer across all variants. Our results show that the hybrid sidecar produces consistent changes in representation geometry despite only marginal aggregate accuracy improvements, with the dominant limitation arising from the restricted representational capacity of the 4-qubit circuit.

2 Related Work

Neural decoding with deep learning. Deep learning models applied to BMI decoding treat neural population activity as spatiotemporal signals [2]. Geometric quality of learned representations—relevant for cross-session robustness and neural drift—is rarely evaluated in this literature.

Quantum machine learning. Variational quantum algorithms and PQCs have been studied as structured function classes [3,8,1]. Quantum feature maps can enhance class separability [5]. Trainability is a central challenge: barren plateaus affect deep circuits [6,4], motivating our use of shallow 4-qubit designs.

Hybrid quantum-classical architectures. Prior hybrid work embeds PQCs as feature-map or kernel layers within classical neural network pipelines [3]. We focus on the residual sidecar setting—a PQC inserted as a differentiable module within a deep neural network—with controlled ablations that analyze geometric effects and gradient connectivity beyond aggregate accuracy.

3 Methods

3.1 Hybrid Architecture

Let $f_{\text{classical}}$ denote ResNet-50 producing feature maps $\mathbf{x} \in \mathbb{R}^{C \times H \times W}$ at layer3 ($C = 1024$). The quantum sidecar is attached here as a residual transform:

$$\mathbf{x}' = \mathbf{x} + f_{\text{quantum}}(\mathbf{x}), \quad (1)$$

and the modified feature map continues through the remaining backbone. We evaluated sidecar placement at layer3, layer4, and the final pooled feature. Layer3

produced the best results on this dataset and model: it ensures the circuit output is processed by subsequent backbone layers, enabling indirect influence on final representations, whereas layer4 or the FC layer restrict the circuit to a last-layer correction with no further backbone co-adaptation.

3.2 Quantum Sidecar

Global average pooling gives $\mathbf{h} = \text{AvgPool}(\mathbf{x}) \in \mathbb{R}^{1024}$. Two learned linear projections map this to circuit angle vectors:

$$\boldsymbol{\theta}_A = \mathbf{W}_A \mathbf{h} + \mathbf{b}_A \in \mathbb{R}^4, \quad \boldsymbol{\theta}_B = \mathbf{W}_B \mathbf{h} + \mathbf{b}_B \in \mathbb{R}^4. \quad (2)$$

Each drives a 4-qubit circuit with a fixed topology (Figure 1). We measure $\mathbb{E}[Z_i]$ for each qubit and project back to $\mathbb{R}^{1024}$.

The **Frozen** variant fixes all projection parameters at random initialization. The **BackboneGrad** variant allows projection parameters to update, but because the quantum simulation is non-differentiable, the input projection receives gradients *only* through the classical backbone pathway—not through the quantum computation itself. Concretely, the gradient path is: Loss $\rightarrow$ FC $\rightarrow$ layer4 $\rightarrow$ layer3 $\rightarrow$ pooled $\rightarrow$ **proj_in**, with no signal flowing from circuit output back to **proj_in**. This means **proj_in** learns to compress features in a way that helps the backbone, but does *not* learn to produce angles that maximize the circuit’s contribution. A separate ablation using the parameter-shift rule provides exact quantum gradients (parameter-shift rule) and resolves this distinction (Section 5.2).

3.3 Circuit Topologies and Mixing

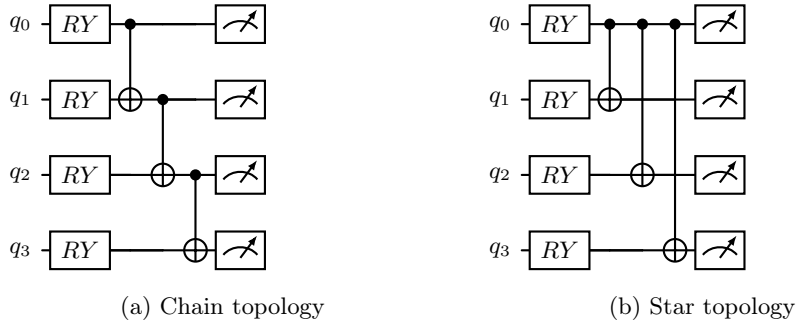


Fig. 1: Circuit topologies (1 of 2 layers shown). Chain: sequential nearest-neighbor CNOTs. Star: qubit 0 as hub with connections to all others.

A soft mixture over both topologies is learned via logits α :

$$f_{\text{quantum}} = \pi_A f_A + \pi_B f_B, \quad \pi_k = \frac{e^{\alpha_k}}{\sum_j e^{\alpha_j}}. \quad (3)$$

All results use noiseless statevector simulation; no quantum advantage is claimed (see Abstract).

4 Experimental Setup

Dataset. 31-class imagined-handwriting neural decoding from multi-neuron spike rasters, subject T5 [2]. Fixed stratified splits: 30,524 train / 3,816 val / 3,816 test samples.

Training protocol. ResNet-50 pretrained on ImageNet; ImageNet-pretrained convolutional features transfer to spatiotemporal spike raster signals as standard practice in BMI deep learning, and training from scratch on 30K samples with a 25M-parameter network would risk overfitting. AdamW with OneCycleLR ($\text{LR}_{\max} = 7 \times 10^{-4}$), batch size 128, 100 epochs, label smoothing 0.05, weighted cross-entropy. Four independent seeds (42, 333, 777, 123). All hyperparameters are held identical across model variants. Measured wall-clock time per epoch (seed 777, single RTX 4090 GPU): Baseline 107s, BackboneGrad 124s, parameter-shift 125s, measurement-guided 124s. All quantum variants add $\approx 16\%$ overhead over the baseline, with the parameter-shift rule adding negligible further cost—4-qubit statevector simulation is fast enough that the 8 circuit evaluations per gradient step are not a bottleneck.

Metrics. Top-1 accuracy, macro-F1, Linear CKA between baseline and quantum feature representations (penultimate layer), the inter/intra cosine centroid ratio $R = \mathbb{E}[\min_{k \neq y} d_{\cos}(\mathbf{x}, \boldsymbol{\mu}_k)] / \mathbb{E}[d_{\cos}(\mathbf{x}, \boldsymbol{\mu}_y)]$, and learned topology mixing weight π_{star} .

5 Results

5.1 Four-Variant Comparison Across Seeds

Table 1 presents the complete multi-seed comparison.

Accuracy. Backbone-gradient training improves accuracy over baseline in 3 out of 4 seeds, with the exception of seed 123 where the coupled optimization found an uncooperative equilibrium. The mean improvement is +0.19% with a 95% CI spanning zero—accuracy differences are not statistically significant at $n = 4$.

Structural reorganization. Linear CKA is strictly lower for BackboneGrad than Frozen in all four seeds (mean $\Delta_{\text{CKA}} = -0.025$, 95% CI $[-0.035, -0.013]$). Unlike accuracy, this directional effect is consistent across every seed, providing evidence of genuine structural reorganization even when accuracy gains are marginal.

Topology preference. Frozen weights remain near chance ($\pi_{\text{star}} \approx 0.504$) in all seeds—the random projection has no topology preference. BackboneGrad weights consistently converge to $\pi_{\text{star}} = 0.579 \pm 0.021$ across all four seeds, revealing a latent star-topology preference that only emerges when the projection co-adapts with the backbone (Figure 3).

Cross-seed variance amplification. BackboneGrad shows $1.66\times$ higher cross-seed accuracy variance than baseline (0.552% vs 0.332%), while Frozen variance is comparable to baseline (0.344%). Frozen simply adds a fixed random projection; its variance reflects a random lottery over projection quality. BackboneGrad introduces joint optimization of backbone and sidecar, creating a more complex loss landscape with additional local minima. Seed 123 illustrates the failure mode, and seed 777 illustrates the best case; the variance is a signature of this sensitivity.

Table 1: Multi-seed comparison of five variants across four seeds. R = inter/intra cosine centroid ratio. CKA = Linear CKA between baseline and quantum features; lower indicates greater structural reorganization. π_{star} = learned star-topology weight at convergence. **BackboneGrad**: input projection trained via backbone pathway only (no gradient through quantum circuit; see Section 3). **Geometry-loss**: CE + feature geometry + angle geometry objectives (see Section 5.2); R column shows the geometry improvement.

Seed	Model	Classification			Geometry ($R \uparrow$)		CKA $\downarrow$	π_{star}
		Top-1	Macro-F1	ΔAcc	Inter/Intra R	ΔR	(vs Base)	(star pref.)
42	Baseline	85.88%	86.18%	—	2.919	—	1.000	—
	Quantum-Frozen	85.53%	85.91%	−0.35%	3.001	+0.082	0.768	0.504
	Quantum-BackboneGrad	85.90%	86.31%	+0.02%	2.959	+0.040	0.747	0.562
	Geometry-loss	85.27%	86.06%	−0.61%	3.809	+0.890	0.577	0.302
	Meas.-guided	85.67%	86.15%	−0.21%	3.823	+0.864	0.613	0.455
333	Baseline	85.22%	85.75%	—	2.897	—	1.000	—
	Quantum-Frozen	86.29%	86.75%	+1.07%	3.054	+0.157	0.743	0.504
	Quantum-BackboneGrad	86.11%	86.66%	+0.89%	3.044	+0.147	0.726	0.575
	Geometry-loss	85.85%	86.88%	−0.26%	3.669	+0.772	0.595	0.574
	Meas.-guided	86.08%	86.63%	−0.03%	3.734	+0.690	0.623	0.462
777	Baseline	85.59%	85.91%	—	2.943	—	1.000	—
	Quantum-Frozen	85.61%	86.00%	+0.02%	3.073	+0.130	0.774	0.504
	Quantum-BackboneGrad	86.32%	86.67%	+0.73%	3.005	+0.062	0.726	0.569
	Geometry-loss	85.59%	86.15%	−0.73%	3.779	+0.836	0.611	0.677
	Meas.-guided	85.67%	86.09%	+0.08%	3.865	+0.860	0.575	0.473
123	Baseline	85.95%	86.75%	—	3.001	—	1.000	—
	Quantum-Frozen	85.72%	85.61%	−0.23%	3.017	+0.016	0.760	0.504
	Quantum-BackboneGrad	85.06%	85.69%	−0.89%	2.942	−0.058	0.745	0.609
	Geometry-loss	85.53%	85.83%	−0.42%	3.894	+0.952	0.610	0.348
	Meas.-guided	85.56%	86.13%	−0.39%	3.775	+0.774	0.609	0.432
Mean (all 4)	Baseline	85.66%	86.40%	—	2.940	—	1.000	—
	Quantum-Frozen	85.79%	86.07%	+0.14%	3.036	+0.096	0.758	0.504
	Quantum-BackboneGrad	85.85%	86.33%	+0.19%	2.988	+0.047	0.736	0.579
	Geometry-loss	85.56%	86.23%	−0.10%	3.788	+0.863	0.598	0.475
	Meas.-guided	85.89%	86.25%	+0.04%	3.799	+0.811	0.605	0.456

Seed 123 anomaly. The implementation of two-phase training—consisting of a 15-epoch sidecar warmup with a frozen backbone followed by joint fine-tuning—was introduced to mitigate the “lazy backbone syndrome” by temporarily stabilizing classical representations during early optimization. However, for problematic random initializations (e.g., Seed 123), this strategy further degraded performance (83.15%, corresponding to a -1.91% drop relative to single-phase training) and produced a highly star-dominant allocation ($\pi_{\text{star}} = 0.848$). These results suggest that effective quantum-classical co-adaptation may require continuous joint optimization from the beginning of training, and that early decoupling of optimization trajectories can bias the system toward unfavorable representational configurations that are difficult to correct during later fine-tuning.

5.2 Ablation Study: Identifying the Bottleneck

We systematically varied nine architectural and training components (Table 2) to identify what limits the quantum sidecar and whether the limitation can be overcome. All variants use seed 777 except geometry-loss and measurement-guided, which were run on all four seeds.

Projection and measurement design. Our first hypothesis was that the simple linear projection from 1024 dimensions to 4 circuit angles might be a bottleneck—either by losing too much information or by mapping all qubits to similar inputs. We tested richer projections (deep MLP, attention) and maximum input diversity (orthogonalized angle subspaces), and also tried full XYZ Pauli readout to increase measurement information. None of these helped. Attention analysis showed $> 99.7\%$ cosine similarity between qubit inputs regardless of projection architecture, and even maximally diverse orthogonal angle inputs produced no improvement.

Classical bottleneck control. A natural sceptical question is whether any benefit comes from the quantum computation itself, or simply from passing features through a low-dimensional bottleneck and back. To test this, we replaced the PQC with an identical $1024 \rightarrow 4 \rightarrow 1024$ architecture using a tanh nonlinearity and the same 18,442 parameters. The classical version actually performs *worse* than the plain baseline, while the quantum circuit holds accuracy above baseline despite the same 4-dimensional compression. The structured entangling gates compensate for compression cost in a way that a simple nonlinearity cannot, ruling out parameter count or bottleneck architecture as explanations for the BackboneGrad improvements.

Gradient connectivity. In the BackboneGrad variant, the input projection receives gradients only through the classical backbone—the quantum circuit itself is not differentiable. One might ask: does the projection actually learn to serve the circuit, or is it just adapting to the backbone regardless? The parameter-shift rule [7] answers this by providing exact quantum gradients through the circuit. With true circuit gradients, projection weights move $184\times$ further from initialization than with backbone gradients alone. Yet the optimizer responds by suppressing the quantum branch ($\gamma \rightarrow 0.003$), meaning the

Table 2: Ablation and variant results (seed 777 unless noted*). Reference: Quantum-BackboneGrad 86.32%. R = inter/intra cosine centroid ratio. Δ = Top-1 relative to reference.

Variant	What changed	Top-1	Δ	R
Quantum-BackboneGrad	Trainable feature-to-angle encoder; backbone-only gradient path; Z-only measurement; learned topology mixing.	86.32%	—	3.005
XYZ measurement	Full $\{X, Y, Z\}$ Pauli readout (12 observables) instead of Z-only (4 outputs).	85.74%	−0.58%	2.902
Deep proj MLP	Shallow linear projection replaced by $1024 \rightarrow 64 \rightarrow 16 \rightarrow 4$ MLP with PCA-initialized output.	85.95%	−0.37%	3.034
Attention projection	Attention-style adaptive projection before angle generation.	85.77%	−0.55%	2.943
Diverse projection	Orthogonality and diversity regularization to decorrelate angle subspaces.	86.08%	−0.24%	3.009
Classical bottleneck	Same $1024 \rightarrow 4 \rightarrow 1024$ residual architecture with tanh nonlinearity; no PQC. Identical parameter count (18,442).	85.14%	−1.18%	2.949
Multiplicative gate	Additive residual replaced by multiplicative modulation: $x \cdot \sigma(\delta)$.	85.77%	−0.55%	—
Parameter-shift	Exact quantum gradients via [7]: $\partial_\theta \langle Z \rangle = \frac{1}{2}[\langle Z \rangle(\theta + \frac{\pi}{2}) - \langle Z \rangle(\theta - \frac{\pi}{2})]$; gradients flow through circuit, not backbone only.	85.40%	−0.92%	2.945
Geometry loss	Explicit feature-space and angle-space geometry objectives added to CE loss ($\lambda_f = 0.03$, $\lambda_a = 0.01$).	85.56%	−0.76%	3.788*
Meas.-guided sidecar	Measurement-derived reward/alignment signal replaces explicit geometry penalty; proj_in adapts to produce circuit-aligned angles.	85.89%	+0.04%	3.799*
U-Net sidecar	Skip-connection encoder-decoder sidecar; larger model capacity requires batch = 64. Reference BG also re-run at batch = 64 (85.88%); Δ is vs that reference. Single seed.	86.45%	+0.57%**	2.945

* 4-seed mean (Table 1).

**U-Net and its BG reference both at batch = 64; not directly comparable to other rows.

best strategy given perfect quantum gradient information is to contribute almost nothing.

Geometry-aware training. Given the task performance ceiling observed under standard cross-entropy, we investigate whether the hybrid architecture can independently improve latent representation geometry—which matters for cross-session robustness [9]? We added explicit objectives penalizing intra-class scatter on the 2048-dim backbone features and enforcing separation in the 4-dim circuit angle space. The geometry improves substantially and reliably across all four seeds ($R = 3.799 \pm 0.082$, 95% CI excluding the BackboneGrad reference of 3.00), but with a small accuracy cost and an important failure mode: the angle-space penalty never decreases below its margin throughout training. The reason is geometric—the kissing-number bound allows only ≈ 24 mutually separated directions in 4D, while we need 31. The circuit simply does not have enough dimensions to separate all classes in angle space, confirming that 4 qubits is a hard capacity ceiling for this task. A second consequence is that geometry pressure erases the spontaneous star-topology preference seen under pure task learning (π_{star} becomes effectively random across seeds), suggesting the star preference depends on classification-driven gradient flow rather than geometric optimization.

Measurement-guided training. The geometry loss showed we can improve R but at an accuracy cost. The measurement-guided variant takes a different approach: instead of imposing an external geometry target, it rewards the projection for producing angle encodings that align with what the circuit actually measures on correctly classified samples. This is a softer, circuit-native signal that does not require specifying a target geometry. Across all four seeds, R improves as consistently as under explicit geometry objectives, while accuracy remains unchanged (mean +0.04%, 95% CI spanning zero). Crucially, the measurement alignment loss actually decreases during training (unlike the geometry angle loss which stalls at its margin), confirming the projection is genuinely learning a compatible encoding that bypasses the rigid dimensional conflict rather than hitting a hard optimization ceiling. The topology preference also shifts: measurement guidance consistently favors chain entanglement over star ($\pi_{\text{star}} = 0.456 \pm 0.017$), the reverse of the spontaneous BackboneGrad preference—suggesting the two circuits discover different but equally valid geometric attractors depending on the training signal.

U-Net sidocar. All previous variants use the same basic sidocar structure. The U-Net variant tests whether additional architectural capacity—skip connections that preserve multi-scale feature information across the bottleneck—changes the picture. Because the larger model requires batch = 64 due to memory, we re-ran BackboneGrad at the same batch size for a fair comparison. U-Net improves accuracy by +0.57% while geometry remains nearly identical to the classical baseline. This suggests that additional hybrid architectural capacity can improve accuracy without substantially altering the underlying representation geometry. Additional multi-seed evaluation is needed to confirm the stability of this effect.

5.3 Per-Class Analysis

Per-class accuracy changes are heterogeneous and class-dependent, and are uncorrelated with class frequency (Figure 2). This rules out the sidcar acting as a class-imbalance correction mechanism and is consistent with a representation-level transformation that alters class manifold geometry in a topology-dependent manner. Row-normalized confusion matrices (not shown) confirm that both models show near-identical strongly diagonal structure, ruling out the sidcar introducing broad misclassification patterns.

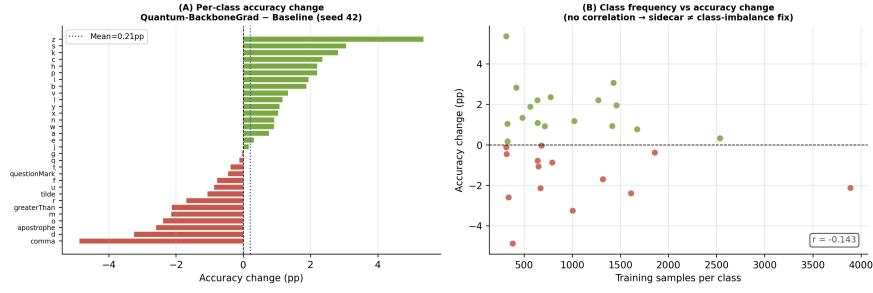


Fig. 2: Per-class accuracy change $\Delta = \text{Acc}_{\text{BackboneGrad}} - \text{Acc}_{\text{baseline}}$ (seed 42). Left: changes are heterogeneous and class-dependent. Right: no correlation with class frequency ($r \approx 0$), ruling out the sidcar acting as a class-imbalance correction.

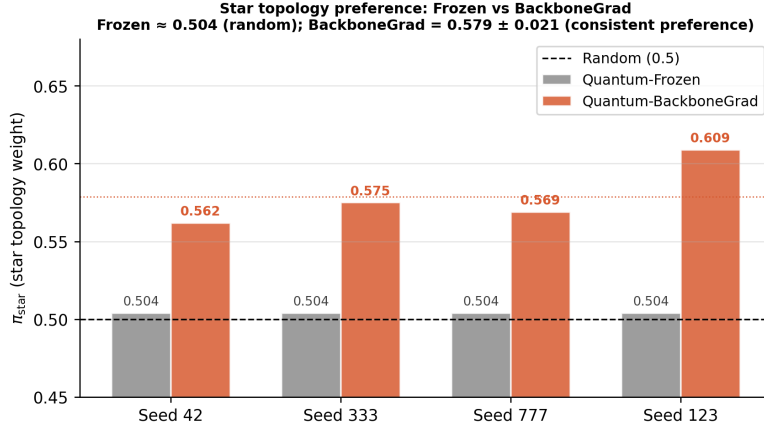


Fig. 3: Star topology mixing weight π_{star} across seeds. Frozen weights remain at ≈ 0.504 (random) in all seeds. BackboneGrad weights converge to 0.579 ± 0.021 in all seeds, revealing a latent preference for hub-and-spoke entanglement.

6 Discussion

Circuit capacity as the binding constraint. The ablation study identifies a single consistent conclusion: the binding limitation is not projection architecture, measurement strategy, gradient connectivity, or input diversity—it is the representational capacity of 4 qubits with 2 entanglement layers producing 4 scalar outputs. The parameter-shift experiment makes this definitive: with exact gradients, $\gamma \rightarrow 0.003$, meaning the optimizer’s best use of the circuit is to contribute almost nothing to a 23M-parameter ResNet. This does not imply that the PQC is inactive—later experiments reveal measurable effects on representation geometry and topology preference—but rather that the current near-term configuration operates in a highly capacity-constrained regime.

Topology preference and neural population coding. The consistent star-topology preference ($\pi_{\text{star}} = 0.579 \pm 0.021$, 4/4 unfrozen seeds) is the most structurally interesting finding in the paper. The star topology places qubit q_0 as a central coordinator whose state directly entangles with all others—a hub-and-spoke structure. The Bolognino et al. dataset used here [2] explicitly demonstrated that motor cortex populations form task-specific coalitions with dominant neurons that coordinate the firing of others, a structure characterized by high betweenness centrality in functional connectivity graphs—precisely the hub topology our model prefers. That four independent random initializations converge to the same topology preference, invisible when the projection is frozen and emerging only through backbone co-adaptation, suggests the model is recovering a structural prior aligned with the population coding geometry of the data rather than a random artifact of training.

Representation geometry, neural drift, and manifold dimensionality. The consistent CKA reduction (4/4 seeds) and geometry ratio improvement provide stronger evidence than accuracy differences, which are statistically underpowered at $n = 4$. The geometry-aware training ablation sharpens this picture across all four seeds: explicitly optimising for class compactness achieves $R = 3.799 \pm 0.082$ (95% CI [3.64, 3.94], excluding the BackboneGrad reference of 3.00) and $\text{CKA} = 0.598 \pm 0.014$ (CI [0.573, 0.623], far below any BackboneGrad value)—both statistically reliable improvements—at a mean accuracy cost of -0.29% versus BackboneGrad, which is not significant. Crucially, the geometry attractor is topology-independent: four seeds produce π_{star} values of 0.677, 0.574, 0.348, 0.302 (mean 0.475 ± 0.155), effectively random, compared to the tight 0.579 ± 0.021 under pure task learning. This dissociation reveals that the spontaneous star preference observed under BackboneGrad is an inductive bias that the circuit discovers through task optimization—not a geometric necessity. This geometry-accuracy dissociation has a direct interpretation for BMI: neural recordings drift across sessions as electrode tuning shifts [9], and models that achieve tight, well-separated class manifolds are more robust to this distributional shift than models optimized for stationary test accuracy alone. Whether the geometry improvements translate to cross-session robustness is the

most important open evaluation, and we flag it as the primary future direction. Cross-session evaluation requires longitudinal intracortical recording data from implanted participants—a structural constraint of the field, as only a very small number of such participants are currently active in research trials worldwide. Separately, the angle geometry loss remaining at its margin value throughout training ($\text{GeomA} = 0.200$, constant) suggests that compressing the 1,024-dimensional backbone representation into only four rotation angles provides insufficient input diversity to reliably organize all 31 character classes within the current encoding scheme. This is consistent with estimates of the intrinsic dimensionality of motor cortex population codes for fine motor tasks, which typically exceed 10–20 dimensions [9]—substantially larger than the 4-angle encoding used here, providing a neuroscientific interpretation of the angle-encoding bottleneck that complements the computational one.

Gradient connectivity and the BackboneGrad distinction. In all experiments except the parameter-shift ablation, the input projection receives gradients only through the classical backbone pathway because the quantum simulation pathway used in the baseline implementation was not part of the differentiable computation graph. The BackboneGrad variant therefore learns feature projections that help the classical backbone adapt around the quantum module, rather than explicitly optimizing angles to maximize the circuit’s own contribution. The parameter-shift experiment (Section 5.2) clarifies this distinction: injecting exact analytic quantum gradients substantially changes the optimization trajectory, yet the circuit’s learned scalar contribution still converges toward near-zero ($\gamma \rightarrow 0.003$). This suggests that the limited task contribution of the current PQC is not solely an artifact of gradient disconnection, but also reflects the restricted capacity of the present low-qubit configuration. For larger hybrid architectures, the distinction between local and global gradient pathways may become increasingly important. Notably, the highly optimized GPU-based state-vector simulation allows exact parameter-shift gradients to be evaluated with relatively small computational overhead in the present setting, strengthening the interpretation that the observed limitation arises primarily from representational constraints rather than optimization instability. This observation motivates the later Geometry-Aware and Measurement-Guided variants, which restructure the training signal to influence representation geometry more directly instead of relying solely on the circuit’s direct discriminative contribution.

7 Conclusion

We evaluated a 4-qubit quantum sidecar in a ResNet-50 backbone under strict experimental controls on 31-class neural population decoding, with a systematic nine-variant ablation study. The main findings are: (1) backbone-gradient-trained input projection consistently causes structural reorganization ($\text{CKA} \downarrow 4/4$ seeds, $\Delta = -0.025$) with directional accuracy improvement (3/4 seeds, CI spans zero), accompanied by $1.66\times$ higher cross-seed variance than baseline,

reflecting the coupled optimization landscape; (2) a latent star-topology preference emerges only with backbone-coupled projection ($\pi_{\text{star}} = 0.579 \pm 0.021$ vs 0.504 ± 0.000 frozen) across all four seeds, consistent with the hub-and-spoke coalition structure of the decoded neural population; (3) the ablation establishes circuit capacity as the binding constraint, with four independent signatures. First, the parameter-shift residual scale converges to near-zero ($\gamma = 0.003$). Second, every alternative projection underperforms the simplest design. Third, geometry-aware training cannot separate 31 classes in 4-dimensional angle space across all four seeds (GeomA stuck at margin = 0.200), consistent with the kissing-number bound of ≈ 24 separable directions in 4D. Fourth, the geometry attractor ($R = 3.799 \pm 0.082$, CKA = 0.598 ± 0.014 , 4-seed 95% CIs both excluding the BackboneGrad reference) is reproducible while topology preference is erased—consistent with the intrinsic dimensionality of motor cortex population codes exceeding the circuit’s 4-dimensional capacity.

These findings do not suggest that hybrid quantum-classical architectures are ineffective—they suggest that this specific combination has reached the ceiling of what a shallow 4-qubit residual sidecar can contribute in the present dataset and architectural regime relative to a strong 23M-parameter backbone. The capacity bottleneck and the gradient disconnection are both architectural, not fundamental, and both point toward the same design question: how should the quantum and classical components be coupled so that the quantum branch is neither suppressed nor redundant? Larger circuits, tighter co-optimization of projection and circuit angles, and training objectives that reward the quantum branch for non-redundant structure are all viable next steps.

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